Ujjaldeep Singh

+91-9873320999 | ujjaldeep.work@gmail.com | LinkedIn | GitHub | Website

Education

Indian Institute of Information Technology, Una — July 2024 – June 2028
B.Tech CSE (Cyber Security); GPA: 8.16 | Branch Silver Medalist for 1st Year — *Himachal Pradesh, India*

Gurukul The School — April 2024
Class XII; 97.40% — *Ghaziabad, U.P.*

Experience

Department for Promotion of Industry and Internal Trade, Ministry of Commerce and Industry, Government of India — June 2025 – July 2025
Software Development Intern — *New Delhi, India*

- Engineered Veridian, a Flask-based password manager supporting encrypted storage of 1k+ credentials.
- Architected authentication workflows with database integration, improving request handling efficiency by 35%.
- Devised AES-based encryption for local storage, ensuring secure protection of all sensitive user data at rest.
- Automated deployment pipelines using Batch and PowerShell scripts, reducing setup time by 60%.
- Optimized frontend interactions, reducing user errors by 25% and improving overall usability.

Projects

NuanceNode – Structured Claim Analysis System | GitHub | Live Demo — March 2026 – Present
Python, FastAPI, Supabase, React.js, REST APIs, Ollama, Redis

- Architected a multi-stage claim analysis pipeline (facts, sources, logic, belief, verdict), enabling structured reasoning across 5 stages.
- Built scalable REST APIs handling 1k+ requests/day with sub-300ms latency for non-AI endpoints.
- Integrated local LLM inference via Ollama, orchestrating 5+ prompt stages, improving response consistency by 30% and reducing token usage by 25%.
- Engineered data layer using SQLAlchemy with MySQL/Postgres, supporting 10k+ records with <100ms query time.
- Enhanced UX with staged loaders, reducing perceived latency by 35% and improving session completion rate by 20%.

NullPass – Passwordless Auth System | GitHub | Live Demo — February 2026
Python, Django, Cryptography, REST APIs, PostgreSQL

- Implemented ECDSA-based challenge–response authentication, eliminating passwords/OTPs and reducing attack surface by 70%.
- Developed secure Django APIs for key registration and signature verification, handling 500+ requests/day with <200ms latency.
- Established device-bound cryptographic identity, reducing credential theft and replay attack risk by 60%.
- Built full-stack authentication flow (React + Django) with client-side key generation, ensuring zero private key exposure.
- Designed zero shared-secret architecture, improving resilience against database leaks and OTP interception by 50%.

Technical Skills

Languages: Python, C/C++, JavaScript
Backend: FastAPI, Django, Flask, Node.js, Express.js, Next.js
Frontend: React.js, Tailwind CSS, Bootstrap, HTML, CSS
Databases: MySQL, PostgreSQL, MongoDB, Redis
Tools: Git, GitHub, AWS, Bash, PowerShell
Security: Cryptography, Authentication, HTTPS/TLS

Positions of Responsibility

Joint-Secretary, EPMOC — December 2025 – Present
IIIT Una